Digital Twin Challenge 2026 by Happiest Health

Rozmarra: A Digital Twin That Predicts Low and High Blood Sugar During Religious Fasting

Rozmarra helps doctors plan safer fasts for people with Type 2 Diabetes in India.

Team NOT_SOMEONE_IMPORTANT, Indian Institute of Technology (Indian School of Mines) Dhanbad

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Rozmarra is a research prototype. It is not a certified medical device, and it must not be used to make decisions about real patients.

Many people with diabetes in India fast, often without medical advice

101 million

people in India have diabetes, and 136 million have prediabetes (ICMR-INDIAB, 2023).

49%

of 300 adults with Type 2 Diabetes in North India had fasted in the previous year (2020 study).

1.35%

of them had discussed their diet with a doctor or dietitian before fasting, and only about 4% had checked their glucose during the fast.

Navratri, Karva Chauth, Ekadashi, Ramadan, Jain fasts and Chhath Puja are a part of everyday life for millions of families. Fasting is a personal and religious decision. Rozmarra aims to help the doctor make each fast safer, and it never judges the decision to fast.

Fasting changes the risk of low and high blood sugar

Medicines

Sulfonylureas and insulin keep lowering glucose when a person does not eat for many hours. A slow CYP2C9 gene variant keeps glimepiride in the body for longer.

Long hours without food or water

Some fasts last from sunrise to moonrise, and some do not allow water. People who take SGLT2 inhibitors may become dehydrated.

Large meals at the end

The meal that breaks the fast is often large and sweet, and it can cause high blood sugar in the hours that follow.

The EPIDIAR study of 12,243 people in 13 countries found that severe low blood sugar needing hospital care was 7.5 times more common during Ramadan for people with Type 2 Diabetes than in other months.

Doctors have a risk score, but it cannot show when and why the risk changes

What doctors have today

- The IDF-DAR 2021 risk score places a person into low, moderate or high fasting risk.
- The IDF-DAR guide asks people to break the fast below 70 mg/dL or above 300 mg/dL, or when they feel unwell.
- Continuous glucose monitors (CGM) and smartwatches record data, but nothing turns it into advice for a fast.

What is missing

- The score is calculated once, so it cannot show the hour of highest risk.
- The doctor cannot test how a lower dose or a new dose time may change the risk.
- No early warning helps during the fast itself.
- Patients rarely receive a written plan in their own language.

Rozmarra supports the doctor before, during and after the fast

Before the fast

The doctor tests a fasting plan on the patient's virtual twin and compares two options, such as a lower dose. Rozmarra shows the glucose curve with its uncertainty band.

During the fast

Rozmarra estimates the chance of low blood sugar in the next 60 minutes from the recent CGM readings, the smartwatch data and the twin. Each alert lists three reasons.

After approval

The doctor approves one plan. Rozmarra prints a plan in English or Hindi with the times to check glucose and the rule for breaking the fast.

Rozmarra never changes medicines by itself. The doctor makes and approves every treatment decision.

Rozmarra combines two data streams

The health record

- Diagnosis, HbA1c, kidney function, weight and blood pressure
- Medicines with doses and times, and the CYP2C9 genotype
- The inputs of the IDF-DAR 2021 score
- Stored as FHIR R4 bundles, and labelled as synthetic

Continuous data

- CGM readings every 5 or 15 minutes
- Heart rate, heart rate variability, steps and sleep from a smartwatch
- Meals, medicine doses and activity in the fasting plan

Data source	Type	Use
ShanghaiT2DM (CC BY 4.0)	Real open data	Fitting the twins and testing predictions on normal days
CGMacros (CC BY-NC-SA 4.0)	Real open data	Fitting the twins and testing the fusion of CGM and smartwatch data
Rozmarra synthetic cohort	Synthetic data	Simulated fasting days of 500 virtual patients for training and testing

Rozmarra supports ten fasting patterns from several faiths

Fasting pattern	Water allowed
Navratri: one meal a day in the evening	Yes
Navratri: fruit and milk during the day, with a vrat meal in the evening	Yes
Karva Chauth: no food or water from sunrise to moonrise	No
Ekadashi: fruit and milk, without grains or beans	Yes
Nirjala Ekadashi: no food or water from sunrise to the next sunrise	No
Ramadan: no food or water from dawn to sunset	No
Jain upvas: no food for the whole day, with boiled water during daylight	Yes
Jain ekasana: one meal a day, before sunset	Yes
Chhath Puja: about 36 hours without food or water	No
Custom fast: the doctor sets every time	Set by the doctor

The fasting times come from sunrise, sunset and moonrise for the patient's city. The patient plan uses the patient's own word for the fast, such as व्रत, उपवास or रोज़ा.

The digital twin models glucose, insulin, meals and medicines

- A physiology model describes glucose, insulin action and insulin for one person, minute by minute.
- Medicine activity curves cover sulfonylureas, premixed and basal insulin, metformin, SGLT2 inhibitors and more.
- The CYP2C9 genotype slows the clearance of sulfonylureas. For example, the *1/*3 genotype gives 65% of the usual speed.
- The team fitted one twin to each of 89 real patients from ShanghaiT2DM and CGMacros.
- 100 parameter samples give a band from the 10th to the 90th percentile.

Forecast error on real held-out days (median RMSE)

Horizon	Twin	Last glucose value
30 minutes	14.5 mg/dL	16.0 mg/dL
60 minutes	23.9 mg/dL	26.8 mg/dL
120 minutes	30.2 mg/dL	38.8 mg/dL

On the Parkes error grid for Type 2 Diabetes, 99.97% of 6,301 forecasts 30 minutes ahead fell in zones A and B, and 97.2% fell in zone A.

The prediction model explains each estimate in plain language

- LightGBM models estimate the chance of low blood sugar (below 70 mg/dL) within the next 60 minutes and the chance of high blood sugar after the fast is broken.
- The features combine CGM readings, 14 days of CGM history, the twin's forecasts, smartwatch signals, the fasting context and the health record.
- Isotonic calibration turns scores into estimated chances.
- A prediction needs a current CGM reading and 70% of the readings of the last 2 hours.

Example alert for Ramesh Patel, Age 62 (synthetic)

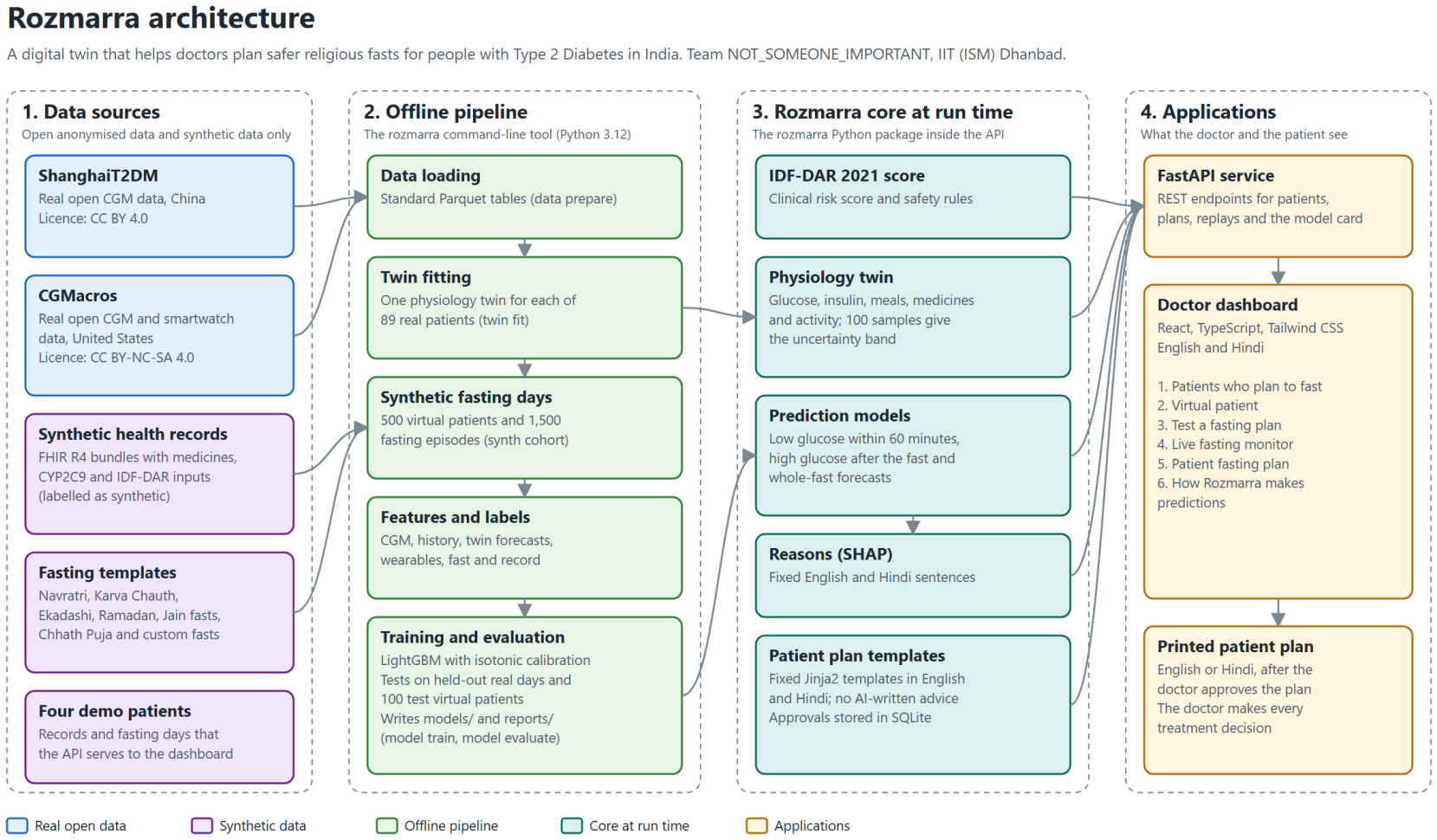
Low glucose is likely within the next 60 minutes.

Estimated chance: 9% (Moderate). Predicted lowest value in the next 2 hours: 74 mg/dL at about 11:04 am.

- The twin predicts a lowest value of 74 mg/dL in the next 2 hours.
- The lowest glucose in the last 2 hours was 77 mg/dL.
- Little food is still being absorbed.

SHAP finds the main reasons, and fixed English and Hindi sentences describe them.

Architecture



All patients in the dashboard are synthetic. The real data comes only from open anonymised datasets, and no raw data is stored in the repository.
Rozmarra is a research prototype that supports the doctor. It is not a certified medical device, and no doctor or pharmacist has reviewed its clinical content.

The team tested Rozmarra on real data and on simulated fasting days

Real normal days

Held-out days of 89 real patients, with five-fold cross-validation grouped by person. These days contain no fasts.

Simulated fasting days

500 virtual patients with 1,500 fasting episodes. The models learned from 400 patients and were tested on 100 patients they had never seen.

Real meals

The model for high blood sugar, trained only on simulated fasts, was applied to 244 real meals that came after at least 8 hours without food.

- The team compared the model with simpler methods: the last glucose value, a straight-line trend, the twin alone, the IDF-DAR score alone and a model that uses only CGM data.
- The 95% intervals come from bootstrapping by patient.
- No open dataset contains CGM data from Indian festival fasts, so the fasting results come from simulation.

Results on real data

Measure	Result	Target	Status
Low blood sugar within 60 minutes, real normal days (AUROC)	0.943 (0.919 to 0.964)	0.80 or higher	Met
Twin forecast error compared with the last glucose value	Lower at 30, 60 and 120 minutes	Lower at every time	Met
30-minute forecasts in zones A and B of the Parkes error grid	99.97%	95% or more	Met
Value of combining data streams, real normal days (AUPRC)	Full model 0.247, CGM only 0.311	Full model higher	Not met
High blood sugar after a real meal that ends a night without food (AUROC)	0.828 (0.725 to 0.903)	No target	Reported

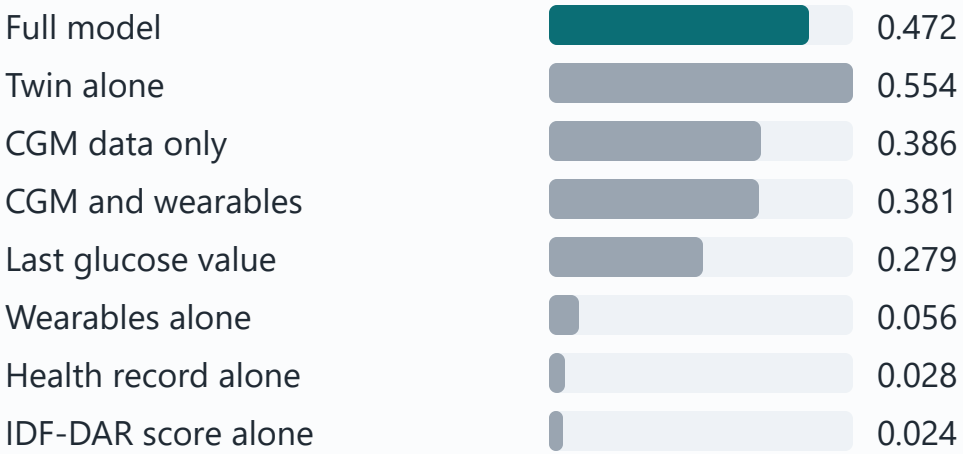
- Only the 14 CGMacros participants have smartwatch data, and they had very few low readings, so the real data does not yet show that smartwatch data adds value.
- On real meals, the glucose value 2 hours before the meal alone reached an AUROC of 0.865, slightly higher than the model.
- The real data comes from China and the United States, not from India.

Results on simulated fasting days

Measure	Result	Status
Low blood sugar within 60 minutes (AUROC)	0.990	Met
Events detected in time (target 80%)	88%	Met
False alerts per patient per fasting day (target 2)	1.9	Met
Median warning time (target 30 minutes)	25 minutes	Not met
High blood sugar after the fast (AUROC)	0.958	Met
Whole-fast forecast of low blood sugar (AUROC)	0.973	Reported

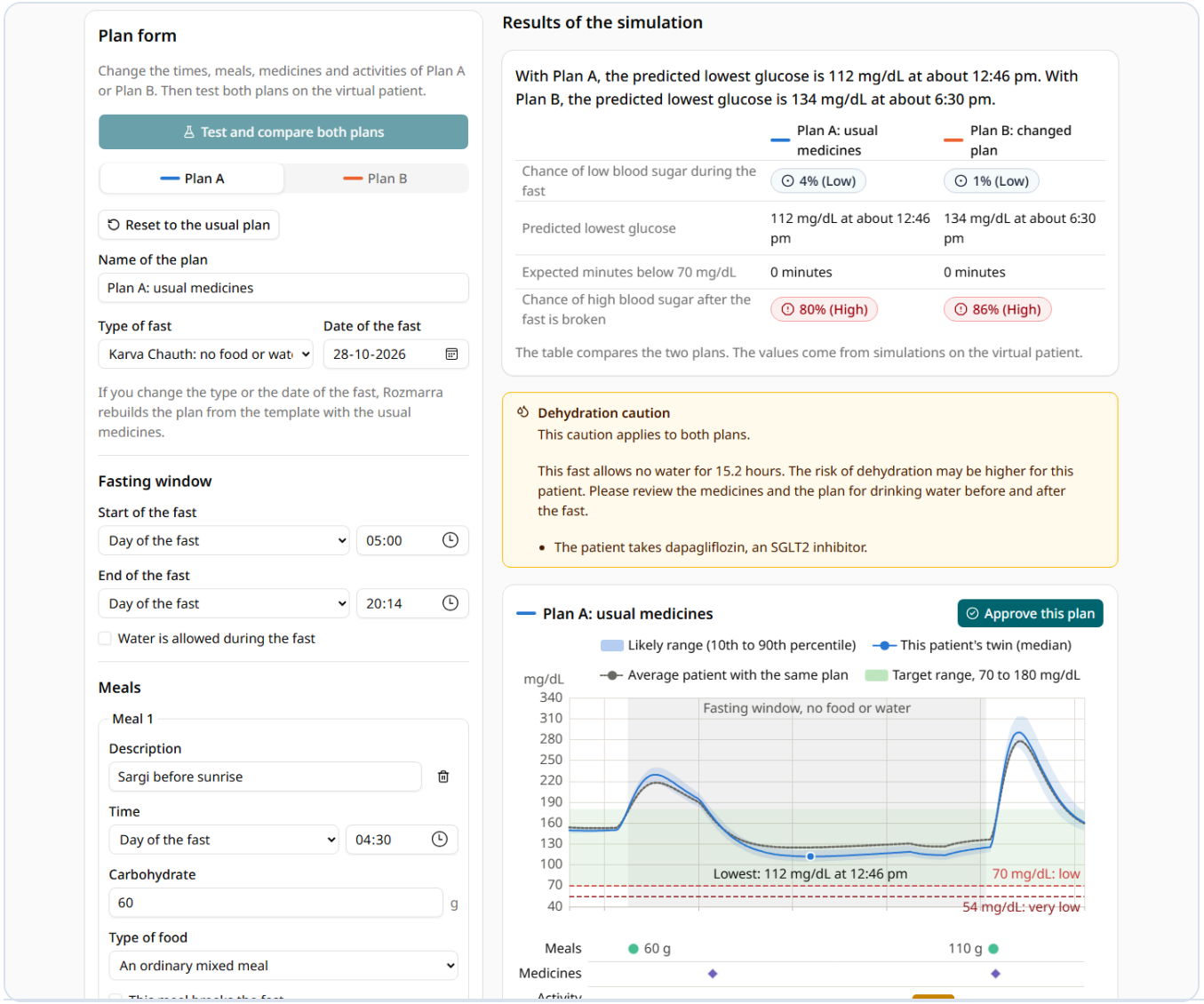
The twin also created the simulated days, so these results partly reflect the team's modelling assumptions.

Ablation study (AUPRC, simulated fasting days)



The full model scored higher than every model that uses one data stream, such as CGM data only or wearables alone. The twin alone scored higher than the full model, partly because the twin created these days.

The doctor tests two fasting plans on the virtual patient

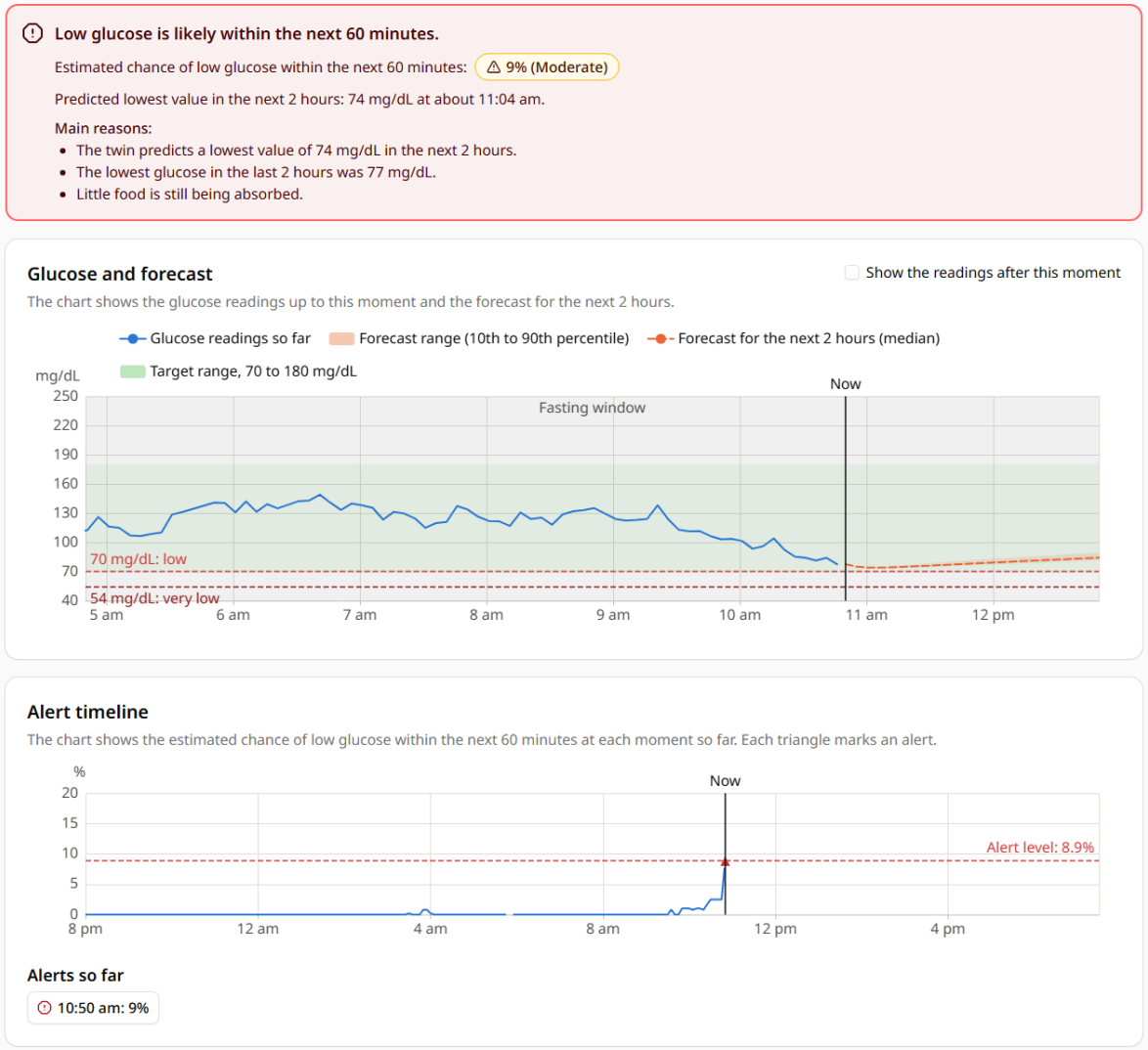


Synthetic patient

Sunita Sharma, Age 54, Karva Chauth

- Plan A, her usual medicines: the lowest glucose is 112 mg/dL at 12:46 pm, with a 4% chance (Low) of low blood sugar.
- Plan B: glimepiride 1 mg with sargi, and dapagliflozin after the fast. The lowest glucose is 134 mg/dL at 6:30 pm, with a 1% chance (Low).
- The dehydration caution appears, because the fast allows no water for 15.2 hours and she takes dapagliflozin.
- The grey curve shows an average patient with the same plan, to show why her personal twin matters.

The live fasting monitor replays a fasting day

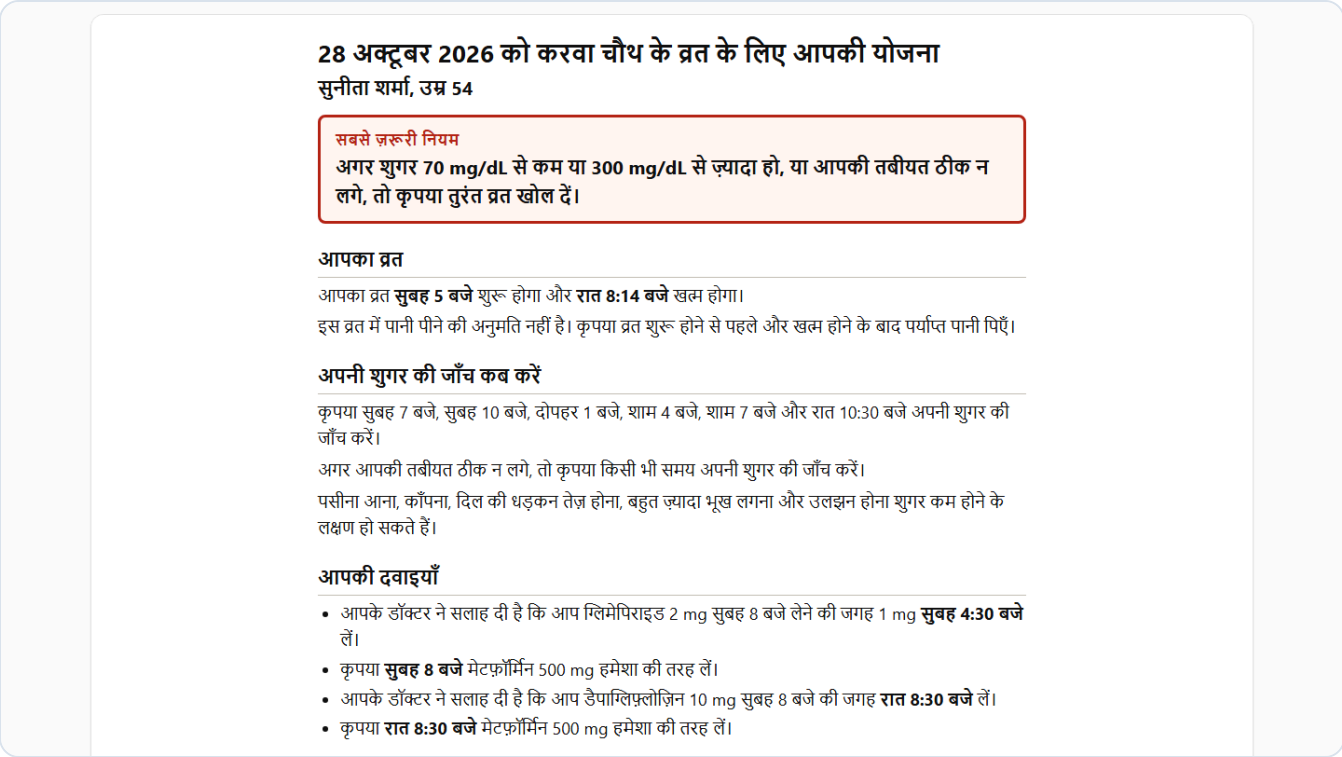


Synthetic patient

Ramesh Patel, Age 62, Navratri

- A time slider moves through the recorded day. At each moment, Rozmarra uses only the data recorded up to that moment.
- The first alert comes at 10:50 am, 41 minutes before glucose first falls below 70 mg/dL at 11:31 am.
- The chart shows the readings so far, the twin's forecast for the next 2 hours and the alert timeline.
- An alert is not repeated within 30 minutes unless the chance rises by 10 percentage points.

The patient receives a printed plan in English or Hindi



28 अक्टूबर 2026 को करवा चौथ के व्रत के लिए आपकी योजना

सुनीता शर्मा, उम्र 54

सबसे ज़रूरी नियम

अगर शुगर 70 mg/dL से कम या 300 mg/dL से ज़्यादा हो, या आपकी तबीयत ठीक न लगे, तो कृपया तुरंत व्रत खोल दें।

आपका व्रत

आपका व्रत **सुबह 5 बजे** शुरू होगा और **रात 8:14 बजे** खत्म होगा।

इस व्रत में पानी पीने की अनुमति नहीं है। कृपया व्रत शुरू होने से पहले और खत्म होने के बाद पर्याप्त पानी पिएँ।

अपनी शुगर की जाँच कब करें

कृपया सुबह 7 बजे, सुबह 10 बजे, दोपहर 1 बजे, शाम 4 बजे, शाम 7 बजे और रात 10:30 बजे अपनी शुगर की जाँच करें।

अगर आपकी तबीयत ठीक न लगे, तो कृपया किसी भी समय अपनी शुगर की जाँच करें।

पसीना आना, काँपना, दिल की धड़कन तेज़ होना, बहुत ज़्यादा भूख लगना और उलझन होना शुगर कम होने के लक्षण हो सकते हैं।

आपकी दवाइयाँ

- आपके डॉक्टर ने सलाह दी है कि आप ग्लिमेपिराइड 2 mg सुबह 8 बजे लेने की जगह 1 mg **सुबह 4:30 बजे** लें।
- कृपया **सुबह 8 बजे** मेटफ़ॉर्मिन 500 mg हमेशा की तरह लें।
- आपके डॉक्टर ने सलाह दी है कि आप डैपाग्लिफ़्लोज़िन 10 mg सुबह 8 बजे की जगह **रात 8:30 बजे** लें।
- कृपया **रात 8:30 बजे** मेटफ़ॉर्मिन 500 mg हमेशा की तरह लें।

- The doctor approves a plan before Rozmarra can print it.
- The plan shows the rule for breaking the fast near the top: below 70 mg/dL or above 300 mg/dL, or when the patient feels unwell.
- It lists the times to check glucose, the medicine changes, advice on water and the meal that breaks the fast.
- Every sentence comes from fixed templates. Rozmarra does not use AI to write medical advice.
- The Hindi plan uses the patient's own word for the fast and the name in Devanagari.

Safety, ethics, respect for faith and limitations

Safety and ethics

- Rozmarra supports the doctor. It never changes medicines by itself.
- It never judges the decision to fast, and it uses each patient's own word for the fast.
- All patients in the dashboard are synthetic, and the real data comes only from open anonymised datasets.
- Patient plan text comes only from fixed templates.

Limitations

- Rozmarra is a research prototype. It is not a certified medical device.
- No doctor or pharmacist has reviewed the clinical content.
- The real data comes from China and the United States, not from India.
- The fasting results come from simulated fasting days.
- The median warning time of 25 minutes is shorter than the target of 30 minutes.
- The IDF-DAR score was designed for Ramadan, and Rozmarra applies it to other fasts as an approximation.

Next steps

A clinical study

A study with Indian patients during real festival fasts would test the predictions and the alert levels, with clinicians reviewing the content.

ABDM integration

Health records could come from the Ayushman Bharat Digital Mission through FHIR, with the patient's consent.

More fasts and languages

The team would add more regional fasts and plans in more Indian languages, such as Marathi, Bengali and Tamil.

A real product would also need approval as a medical device, for example from the Central Drugs Standard Control Organisation (CDSCO).

Thank you

Team NOT_SOMEONE_IMPORTANT, Indian Institute of Technology (Indian School of Mines) Dhanbad

Member	Role
Shivansh Gautam	Team leader. Leads the digital twin and the prediction models.
Aman Soni	Leads the data loaders, the synthetic data, the API and the dashboard.
Divy Sharma	Leads the fasting templates, the architecture diagram, the presentation, the model card, the data card and testing.

The team thanks the authors of ShanghaiT2DM and CGMacros for sharing open data, and Happiest Health for organising the Digital Twin Challenge 2026. The code is released under the Apache License 2.0.